

Water & Carbon Dioxide Overview

A quick reference guide

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Water (H₂O)

Key Properties

Property	Value
Molecular Formula	H ₂ O
Molar Mass	18.015 g·mol ⁻¹
State at 25 °C	Liquid
Boiling Point	100 °C
Freezing Point	0 °C
Density (20 °C)	0.998 g·cm ⁻³

Water is a polar inorganic compound that is essential for all known forms of life. Its high specific heat capacity and surface tension make it a unique solvent. The hydrogen bonds between molecules give water its anomalous expansion upon freezing.

Carbon Dioxide (CO₂)

Key Properties

Property	Value
Molecular Formula	CO ₂
Molar Mass	44.01 g·mol ⁻¹
State at 25 °C	Gas
Boiling Point	-78.5 °C (sublimes)
Density (1 atm, 25 °C)	1.977 g·L ⁻¹
Solubility in Water	1.45 g·L ⁻¹

Carbon dioxide is a colorless gas produced by combustion and respiration. It is a key greenhouse gas, trapping infrared radiation and contributing to global warming. In aqueous solution it forms carbonic acid, which plays a crucial role in the buffering capacity of natural waters.